Table-top lab · guide

How a Hand Works

Gentle pulls only: no yanking strings or bending fingers backward. Toys differ; a paper-and-straw hand works too.

Try 1 · What keeps a finger stiff?

Gently press along one of your fingers and count the hard parts between the bends. Count the stiff pieces on one toy finger too. Do the two numbers match?

Look for: Bones. Each finger has three small bones in a row, and the thumb has two. Counting the wrist and palm, one hand has 27 bones. Bones hold the shape, but they cannot move by themselves.

Try 2 · What pulls a finger closed?

Draw around your hand on paper to make an outline. Pull one toy string slowly and watch which finger curls. Draw a line from that string to the same finger on your outline. Do the same for every string, one at a time.

Look for: In the toy, a string. In your hand, a tendon: a tough cord that joins a muscle to a bone. Muscles in your forearm pull long tendons that reach into each finger, and tough loops along the finger hold them close, like the toy's guides.

Try 3 · Does your hand work like the toy?

Curl your pointer finger slowly and count the places it bends. Curl the toy's pointer finger and compare. Rest your other hand on your forearm and wiggle your fingers. What do you feel move?

Look for: Both follow one plan: a pull, a string, and joints that bend. Your finger bends at three joints, mostly one way, like a door hinge. The muscles that curl your fingers sit mostly in your forearm, so you can feel them move there.

Try 4 · What opens the hand again?

Let go of a toy string and watch whether the finger springs back. Look for what pulls it open. Then lay your hand flat, fold your middle finger under, and gently try to lift your ring finger.

Look for: A muscle can only pull, never push. So a second set of tendons runs along the back of each finger and pulls it straight. Many toys use an elastic cord or a spring instead. Some tendons on the back of your hand are linked, which is why one finger can be hard to lift alone.

Print the lab sheet and this guide separately.